

The sintering-timescale map (Figure 8): what goes into it

enceladus_DSM, studies/keff_sintering/timescale_map/

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Summary

Figure 8 is **not simulation output**. It is one closed-form expression,

$$t(T, R) = \theta^* \tau_{\text{sub}}(T, R), \quad (1)$$

evaluated on a grid of temperature T and grain radius R . One number comes from the campaign, the anchor $\tau_{\text{sub}} = 7822.3$ s at -20°C and $R = 50$ μm ; one result of the campaign justifies the form, the temperature collapse; everything else is the model's own definition of τ_{sub} , written in the quantities the model uses. Only one grain size and temperatures from -40 to -5°C were simulated, so every other point of the map is an extrapolation. Section 7 checks the annotations against the two framing papers and Section 8 lists what the extrapolation assumes. Every number below is recomputed by `check_timescale_map.py` in this folder (output in `check_timescale_map.txt`).

1 The phase-change time scale

The model defines (solver banner; material-properties table of the manuscript)

$$\tau_{\text{sub}} = \frac{\varepsilon^2 \beta_{\text{sub}}}{d_0}, \quad \beta_{\text{sub}} = \frac{\rho_{\text{ice}}}{\rho_{vs}^I(T)} \frac{1}{\alpha_c} \sqrt{\frac{2\pi m}{k_B T}}, \quad d_0 = \frac{\gamma a^3}{k_B T}, \quad a^3 = \frac{m}{\rho_{\text{ice}}}, \quad (2)$$

with ε the interface parameter, α_c the condensation coefficient, m the mass of a water molecule, γ the ice–air interfacial energy and ρ_{vs}^I the saturation vapor density over ice. Substituting,

$$\boxed{\tau_{\text{sub}} = \frac{\varepsilon^2 \rho_{\text{ice}}^2}{\alpha_c \gamma m} \frac{\sqrt{2\pi m k_B T}}{\rho_{vs}^I(T)}} \quad (3)$$

Equation (3) reproduces the solver's τ_{sub} at all five simulated temperatures to within 0.03% (60 389 against 60 390 s at -40°C ; 7822 against 7822 s at -20°C).

2 Temperature dependence

In Eq. (3) the temperature enters only as

$$\tau_{\text{sub}} \propto \varepsilon^2 \frac{\sqrt{T}}{\rho_{vs}^I(T)}, \quad (4)$$

and the map evaluates exactly this, with $\rho_{vs}^I(T)$ the model’s own function, the expression the solver uses (`rho_vs_sat` in `comp_eps.py`). No saturation pressure appears: the model has no pressure variable, and none is needed. At the five simulated temperatures the map’s τ_{sub} therefore equals the solver’s (Table 1); away from them it is the same function evaluated at other temperatures.

Table 1: The map’s τ_{sub} against the solver’s, at $R = 50 \mu\text{m}$ ($\varepsilon = 1 \mu\text{m}$).

T	τ_{sub} map (s)	τ_{sub} solver (s)	difference
-40°C	60 389	60 390	0.00%
-30°C	20 835	20 840	−0.03%
-20°C	7 822	7 822	anchor
-10°C	3 164	3 164	0.00%
-5°C	2 063	2 063	+0.02%

Effective activation energy. $E = -R_g \text{d} \ln[\rho_{vs}^I/\sqrt{T}]/\text{d}(1/T) = 50.1 \text{ kJ mol}^{-1}$ at 253 K and 50.2 at 180 K. The same number follows from the solver’s values alone: $\ln(60\,390/2063)$ over the simulated range gives 50.1 kJ mol^{-1} . *This corrects the “about 48 kJ mol^{-1} ” quoted earlier in the project notes*, which came from an ideal-gas form of ρ_{vs}^I and not from the model’s.

Range of $\rho_{vs}^I(T)$. The model’s ρ_{vs}^I rests on a polynomial fit for the saturation state over ice. As recalled, such fits are stated for about -100 to 0°C (*to verify*); the map uses the function down to 50 K. It stays smooth and monotonic there, but below roughly 173 K it is the fit extrapolated, not a tabulated property.

3 Grain-size dependence

There is **no grain-size data** in the figure. The vertical axis is the variable R in Eq. (1); only $R = 50 \mu\text{m}$ was simulated.

The R^2 law must not be argued from $\varepsilon \propto R$. The campaign sets $\varepsilon = \bar{R}/50$. That is a numerical choice (it fixes the mesh at any grain size), and $\tau_{\text{sub}} = \varepsilon^2 \beta_{\text{sub}}/d_0$ is the relaxation time of the diffuse interface, not a physical time. Reading “ $\tau_{\text{sub}} \propto \varepsilon^2 \propto R^2$ ” as the grain-size law would make the physics follow from a discretization rule. It does not, and the interface-width runs show it directly: on three packings at fixed $R = 50 \mu\text{m}$, doubling ε (so τ_{sub} goes from 7822 to 31 289 s, a factor 4.00) leaves the kinetics *in physical time* unchanged after the early transient. From day 8 to day 30, k_{eff} rises 8.1, 9.4 and 8.8% at $\varepsilon = 1 \mu\text{m}$ and 8.0, 9.6 and 8.7% at $2 \mu\text{m}$, and SSA falls 10.2–11.2% against 10.1–10.3%. If the rate followed $1/\tau_{\text{sub}}$ the coarser runs would be four times slower; they are not. So at fixed R the physical rate is independent of ε , as it should be for a phase-field model near its sharp-interface limit, and τ_{sub} by itself carries no grain-size information.

Where R^2 does come from: the sharp-interface law. In the limit the model is built to reproduce (Kaempfer & Plapp, 2009), the interface moves with normal velocity

$$\beta_{\text{sub}} v_n = u - d_0 \kappa, \quad (5)$$

with u the dimensionless supersaturation at the interface and κ the curvature. When attachment limits the rate, u is uniform over the pore and set by conservation of mass, so $v_n = (d_0/\beta_{\text{sub}})(\bar{\kappa} - \kappa)$: motion by curvature at fixed volume. Measuring lengths in R ($\kappa \sim 1/R$, $v_n \sim R/t$) gives the only time scale in the problem,

$$\tau_R \equiv \frac{R^2 \beta_{\text{sub}}}{d_0}, \quad (6)$$

and no ε appears. Two packings that differ only by a uniform change of scale therefore evolve identically in t/τ_R . This is the grain-size law of the map. Because the campaign happens to use $\varepsilon = R/50$, the two times are related by $\tau_{\text{sub}} = \tau_R/2500$, and that is the *only* reason the map can be written with τ_{sub} ; the physical statement is $t \propto \tau_R$.

What this rests on.

- *ε -converged kinetics.* Tested once, towards a coarser interface ($\times 2$, three packings): late-time rates agree as above; absolute k_{eff} differs by 1–3% and the first day or so is ε -dependent. Not tested towards a finer interface ($\varepsilon = 0.5 \mu\text{m}$ is an optional run in `TODO.md`).
- *Attachment-limited transfer.* Equation (5) with uniform u needs $R \ll L^*$ (Section 8); otherwise vapor diffusion adds a second time scale and the law steepens towards R^3 .
- *Geometric similarity:* the same packing, scaled.
- *No run at a second grain size.* The law is derived, not measured, by us. It agrees with the grain-size dependence in Molaro et al. (2019); see Section 7.

Consequence for the sintering age. $\theta = t/\tau_{\text{sub}}$ is defined with an ε -dependent time. It is a valid clock across temperature because every run uses the same ε and β_{sub}/d_0 carries the temperature. It is not transferable as a number: the same microstructural state that sits at $\theta = 331$ here sits at $\theta = 83$ in the $\varepsilon = 2 \mu\text{m}$ runs. The ε -free age is $t/\tau_R = \theta/2500$.

4 Why time scales with τ_{sub} at all

Equation (1) assumes the microstructure is a function of the sintering age $\theta = t/\tau_{\text{sub}}$ alone. Across temperature this is the campaign’s main result (Figure 6): for each of the 25 packings, runs at five temperatures give the same k_{eff} at matched SSA to within 0.27% (100 pairs; median 0.07%), and the time needed to reach a given state follows the ratio of τ_{sub} to within 4.1%. So within -40 to -5°C the statement “time to a given state $\propto \tau_{\text{sub}}$ ” is measured, to 4%.

5 The target age θ^*

$\theta^* = 331$ is the age the -20°C runs reach in 30 days: $30 \text{ d}/7822.3 \text{ s} = 331.4$. It has no other significance; any age gives the same map shifted by a constant factor. In the results, at $\theta = 331$ the 15 well-connected packings (porosity ≤ 0.375) have

$$k_{\text{eff}}/k_{\text{eff},r} = 1.171 \pm 0.015, \quad \text{SSA}/\text{SSA}_r = 0.819 \pm 0.007$$

relative to the reference age $\theta_r = 30$ (mean and standard deviation over packings). The map is therefore “time for k_{eff} to rise 17% and SSA to fall 18% from their values at θ_r ”.

6 The formula as plotted

$$t(T, R) = \theta^* \tau_{\text{sub}}^{\text{ref}} \frac{\sqrt{T} / \rho_{vs}^I(T)}{\sqrt{T_{\text{ref}}} / \rho_{vs}^I(T_{\text{ref}})} \left(\frac{R}{R_{\text{ref}}} \right)^2, \quad (7)$$

with $\theta^* = 331$, $\tau_{\text{sub}}^{\text{ref}} = 7822.3$ s, $T_{\text{ref}} = 253.15$ K and $R_{\text{ref}} = 50$ μm . It is evaluated for $50 \leq T \leq 260$ K and $0.1 \mu\text{m} \leq R \leq 1$ mm and binned at 1 h, 1 d, 1 yr, 1 kyr, 1 Myr and 4.5 Gyr (`fig.timescales` in `figures/sample_figures.py`).

Table 2: Values of the map. Only the first row, last column, is a simulated condition.

T (K)	$R = 1 \mu\text{m}$	$5 \mu\text{m}$	$6 \mu\text{m}$	$50 \mu\text{m}$
253.15	17 min	7.2 h	10.4 h	30 d
220	10.4 h	10.9 d	15.7 d	3.0 yr
200	6.8 d	169 d	244 d	46 yr
180	194 d	13.2 yr	19.1 yr	1.3 kyr
150	428 yr	10.7 kyr	15.4 kyr	1.1 Myr
120	9.4 Myr	235 Myr	339 Myr	23.5 Gyr
110	873 Myr	21.8 Gyr	31.4 Gyr	> age of the Solar System
80		> 10^{17} yr at every grain size		

7 The annotations, checked against the literature folder

All four annotations are supported by the two framing papers in `Literature/`. One bracket was wrong and is corrected in the figure.

What Choukroun et al. say about grain size. Not a measurement: they used one grain size. Their statement (Sect. 5, reason 4 for extrapolating) is that “the effect of temperature on sintering rates strongly dominates over that of grain size (Molaro et al., 2019), such that a 1 to 2 order of magnitude difference in the mean grain size between our samples and actual fresh plume deposits ... is not anticipated to alter the conclusions of this study.” The map agrees on which variable dominates (temperature moves the time by more than 30 orders of magnitude across the plot, grain size by 8), but under an R^2 law one to two orders of magnitude in grain size is two to four orders of magnitude in time, which is not negligible for a stated timescale.

What Molaro et al. give for grain size. Their Table 6 fits the stage-1 sintering timescale as $\tau = f T^g$ for radii from 0.1 to 100 μm . From those fits, the local exponent $d \ln \tau / d \ln R$ is 2.00 between 50 and 100 μm , 2.01 between 30 and 50, 1.95–2.02 between 10 and 30, and 1.7–2.3 between 1 and 10 μm , at 130–180 K. Their model, which includes surface diffusion and is independent of ours, therefore also gives $\tau \propto R^2$ above about 10 μm . This is the best available support for the map’s grain-size axis.

Annotation	Value plotted	What the papers say
“plume grains”	$R = 0.1\text{--}5\ \mu\text{m}$	Choukroun et al. (2020), Sect. 1: “Particles with radii $0.1\text{--}5\ \mu\text{m}$ are expected to dominate the plume deposits” (citing Kempf et al., 2010; Southworth et al., 2019); mean radius $3.1 \pm 0.5\ \mu\text{m}$ (Ingersoll & Ewald, 2011). Molaro et al. (2019), Sect. 5.3: plume fallout of $0.25\text{--}2.5\ \mu\text{m}$ radius (Kempf et al., 2010). Confirmed.
“surface”	50–80 K (<i>was 60–80</i>)	Choukroun, Sect. 5: “south polar terrain mean temperatures can be colder than 50 K, while noontime equatorial temperatures can reach over 80 K” (Howett et al., 2010). Molaro, Sect. 5.4: maximum equatorial surface temperature of about 80 K (Howett et al., 2010). Lower end corrected to 50 K.
“Tiger Stripes”	180 K (<i>was a 175–185 K bracket labelled “fractures”</i>)	Choukroun, Sect. 5: “temperatures up to 180 K have been estimated” (Spencer & Nimmo, 2013). A single value, so it is now a tick, not a range.
Choukroun marker	180 K, $R = 6\ \mu\text{m}$, 15 yr	Choukroun, Sect. 2: “ $12\ \mu\text{m}$ mean diameter particles”, porosity $51.5 \pm 1.6\%$, sealed, 193–243 K; Sect. 5: about 10 MPa “in ~ 15 years” near the Tiger Stripes. Confirmed.

Their timescales against the map. At 180 K Molaro’s fit gives 2.3 d, 126 d and 8.2 yr for $R = 1, 10$ and $50\ \mu\text{m}$; the map gives 194 d, 53 yr and 1.3 kyr. The two are not the same quantity (theirs is completion of stage 1 with all mechanisms; ours is a 17% rise in k_{eff} by vapor transport), but the direction is informative: their full model is about 100 times faster than our vapor-only clock at this temperature, as expected if surface diffusion carries cold sintering.

8 What the extrapolation assumes

1. **Temperature range.** Simulated: 233–268 K. Plotted: 50–260 K. The collapse is measured over 35 K and applied over 210 K.
2. **One grain size.** The R^2 law is a scaling argument, supported by Molaro’s Table 6 but not by a run of ours.
3. **Attachment-limited transfer everywhere.** The crossover length between attachment and diffusion control is $L^* = D_v \beta_{HK}$, $\beta_{HK} = \alpha_c^{-1} \sqrt{2\pi m / (k_B T)}$: $139\ \mu\text{m}$ at 253 K and $89\ \mu\text{m}$ at 180 K for air at 1 atm and $\alpha_c = 10^{-3}$. The R^2 law needs $R \ll L^*$. The simulated $R = 50\ \mu\text{m}$ is already at $R/L^* = 0.36$, and the upper third of the map ($R > 100\ \mu\text{m}$) is beyond it; there diffusion slows growth and the map *underestimates* the time.
4. **Constant** $\alpha_c = 10^{-3}$ at all temperatures. Its value near 100 K is not known to us.
5. **Air-filled pores at 1 atm.** The pore gas enters through D_v and L^* . Near the surface of Enceladus the pores are close to vacuum; Choukroun et al. make the same point about their sealed samples.

6. **Vapor route only.** Surface diffusion is absent from the model. Molaro et al. and Choukroun et al. both attribute cold sintering to it, the latter with $Q = 24.3 \pm 3.3 \text{ kJ mol}^{-1}$ against our 50, and against the $\sim 51 \text{ kJ mol}^{-1}$ they quote for vapor diffusion. A missing mechanism can only make sintering faster, so the map is an *upper bound on the time*: the time the vapor route alone would take.
7. $\rho_{vs}^I(T)$ **below its fitted range** (Section 2).
8. **The age–state relation** (SSA linear in $\log \theta$) is itself an intermediate-time approximation; the map extrapolates the clock, not that relation.

The Choukroun point. The map gives 19 yr at 180 K and $R = 6 \text{ }\mu\text{m}$; Choukroun et al. give about 15 yr. These are not the same quantity. Theirs is the time to a strength of 10 MPa by a mechanism they identify as surface diffusion; ours is the time to an arbitrary sintering age by vapor transport. The closeness of the two numbers is not evidence that the model reproduces their experiment.

9 Reproducing it

```
cd studies/keff_sintering
python timescale_map/check_timescale_map.py <campaign dir>
python figures/sample_figures.py <campaign dir> --skip-gallery
# the second writes Figure8_timescales; use venv_enceladus/bin/python
```